Howe truss

Plane truss — analysis and CIRSOC 301 rod checks

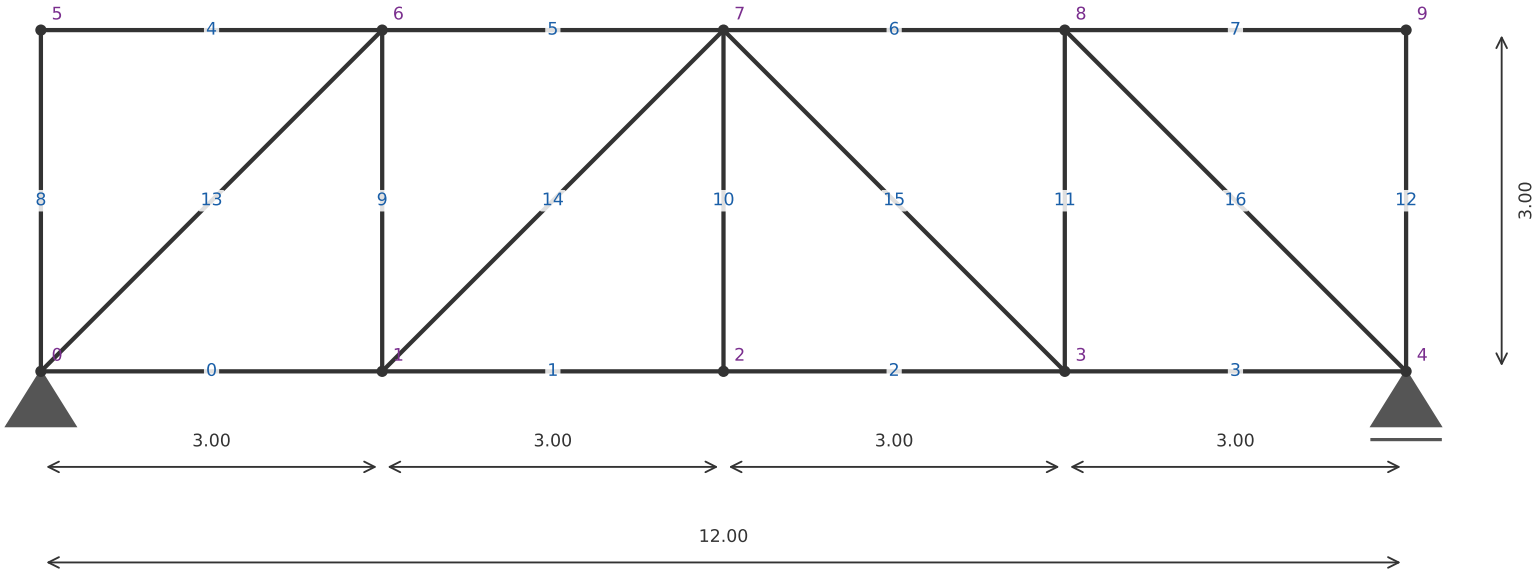
Span	12.00 m
Height	3.00 m
Nodes / rods	10 / 17
Supports	pin at 0, rollerX at 4
Loads	3 nodal, total 120.0 kN
Family "Chord"	CHS 114.3x4.5 (F24 (CIRSOC) / A36)
Family "Diagonal"	CHS 101.6x4.0 (F24 (CIRSOC) / A36)
Family "Vertical"	CHS 60.3x3.6 (F24 (CIRSOC) / A36)
Weight	Weighs 542 kg, carries 120.0 kN (23 × its own weight), passes (0.69).
Deflection	5.94 mm = L/2020, limit L/300: OK
Stability	Statically determinate (m + r − 2j = 17 + 3 − 2×10 = 0): statics alone gives every force.
Most used rod	rod 13 at 0.69 (inelastic buckling (E3-2))
Rods over capacity	0 of 17 checked
Buckling	$\lambda_1 = 1.90$
Shear and moment	none: every rod is pin-end loaded, so the axial force is the whole result

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9. Explain rod 13 and 16

Results are an aid to a qualified professional who verifies them. Rods checked to CIRSOC 301 assuming pin-ended first-order analysis; joints and connections are not checked here.

Node numbers in purple, rod numbers in blue. Dimensions in m.



2 m · true to scale

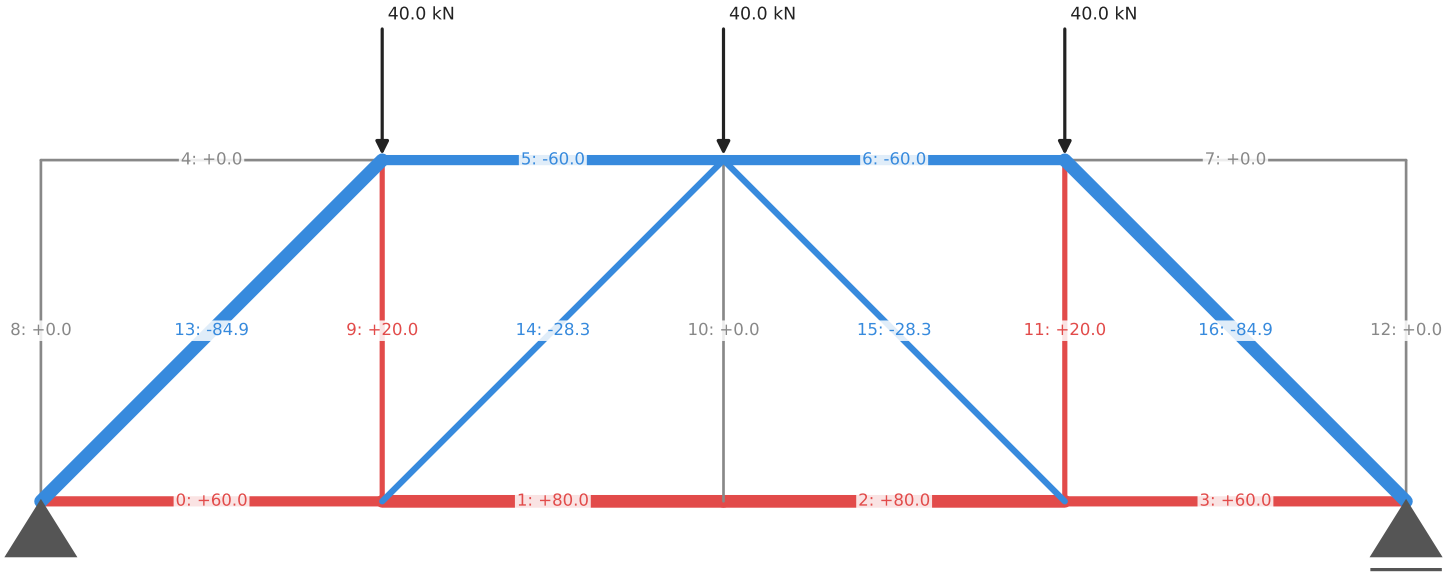
0 2

Truss — Geometry

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MODEL	SIZE	UNITS	CODE	DATE
Howe truss	10 nodes / 17 rods	m, kN, mm	CIRSOC 301	2026-10-12 01:58

Axial force N (kN): red pulls (+), blue pushes (−), grey carries nothing; line width follows |N| (largest 84.9).



2 m · true to scale

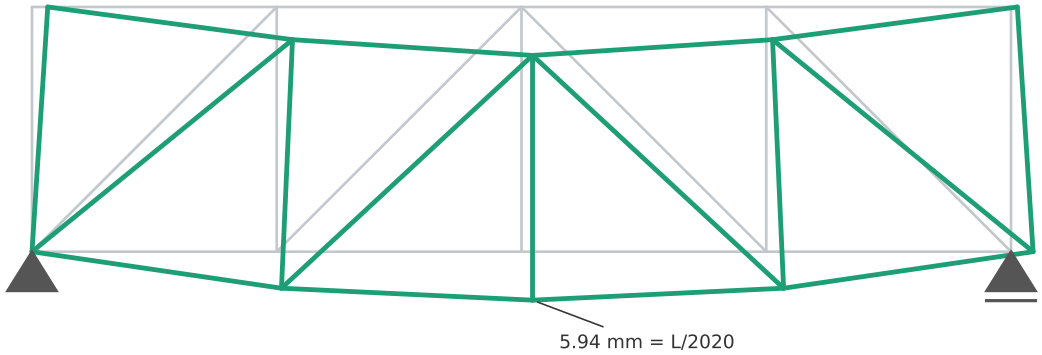
0 2

Truss — Axial forces

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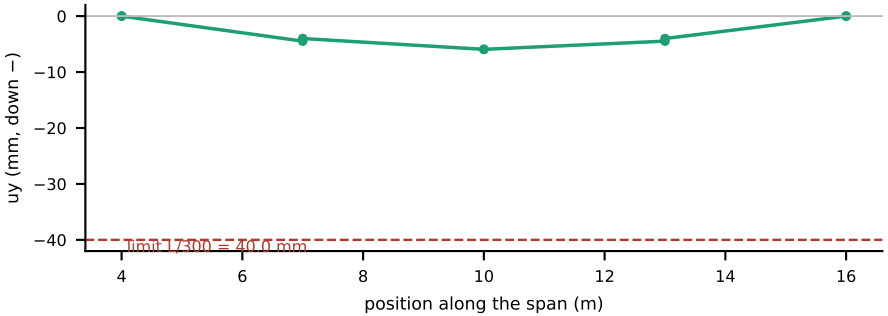
MODEL	SIZE	UNITS	CODE	DATE
Howe truss	10 nodes / 17 rods	m, kN, mm	CIRSOC 301	2026-10-12 01:58

Deflected shape (green) over the truss (grey). Geometry to scale; displacements magnified × 100. Largest vertical deflection 5.94 mm against the limit L/300 = 40.0 mm: OK.



Elastic deformation

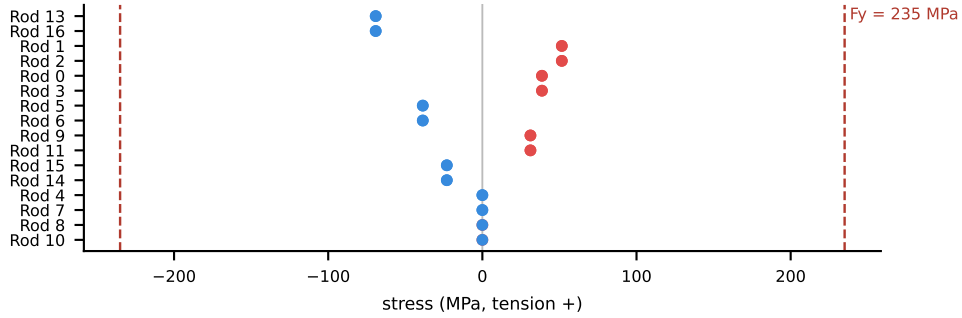
vertical movement of every joint



Largest 5.94 mm at joint 2 = L/2020. Within the limit.

Fibre stress

N/A ± M·c/I at each rod's extreme fibres



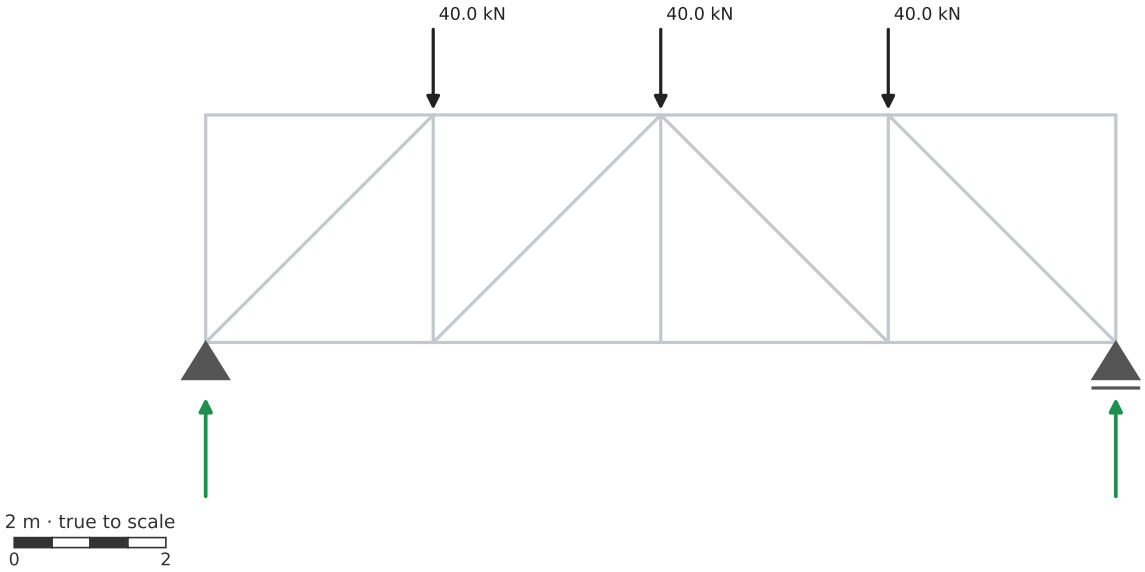
Every rod is pinned, so its two fibres carry the same N/A and coincide: no bending opens the pair.

Truss — Deflection and stress

MODEL	SIZE	UNITS	CODE	DATE
Howe truss	10 nodes / 17 rods	m, kN, mm	CIRSOC 301	2026-10-12 01:58

Support reactions (green arrows), as the supports push on the truss; Ry positive upwards. The sums balance the loads.

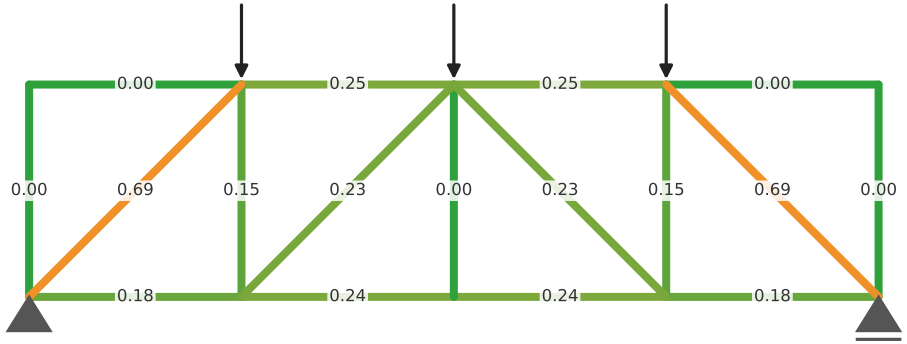
Node	Type	Rx (kN)	Ry (kN)
0	pin	0.00	60.00
4	rollerX	0.00	60.00
Σ		0.00	120.00



Truss — Reactions

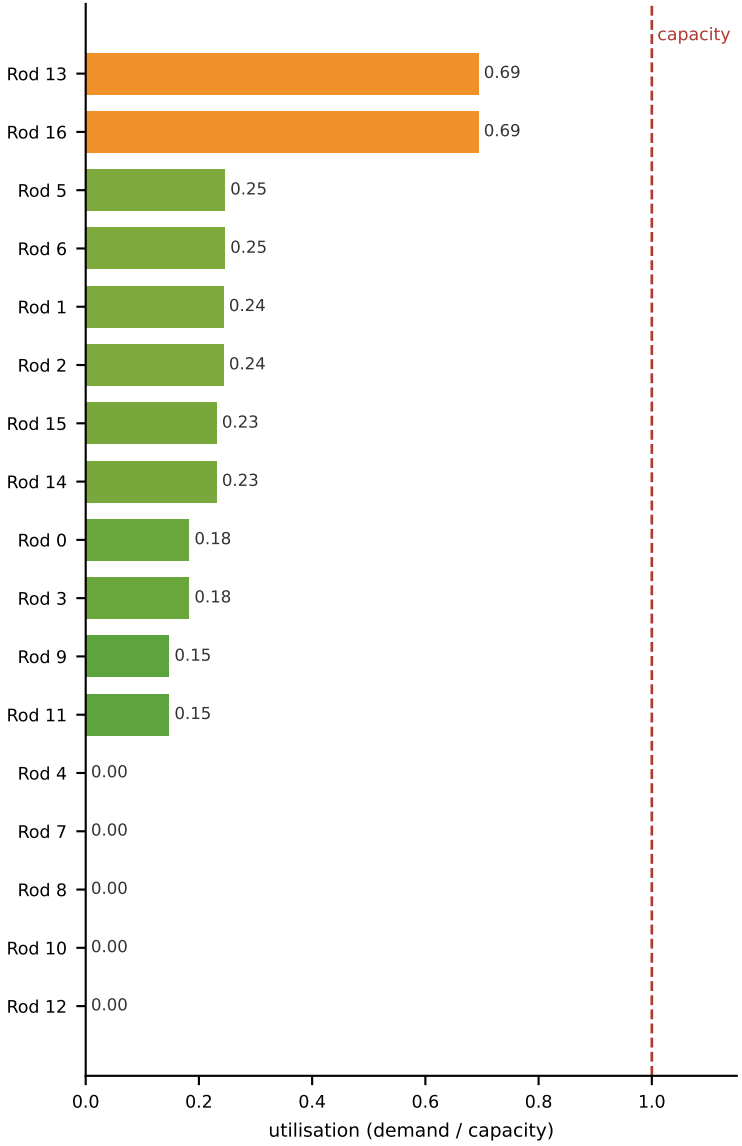
MODEL	SIZE	UNITS	CODE	DATE
Howe truss	10 nodes / 17 rods	m, kN, mm	CIRSOC 301	2026-10-12 01:58

Utilisation: demand over CIRSOC 301 capacity. Most used rod 13 at 0.69 (inelastic buckling (E3-2)). Every checked rod is within capacity.



2 m · true to scale

0 2





grey = no steel section, not checked

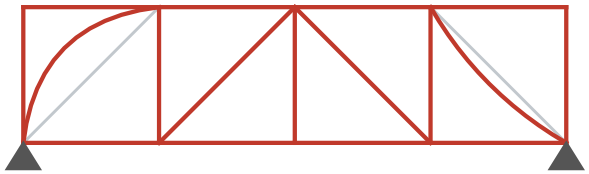
Truss — Utilisation

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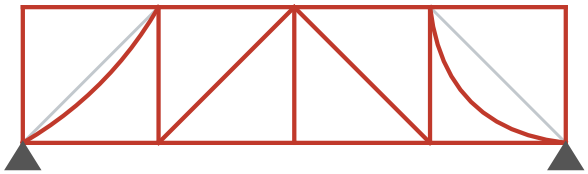
MODEL	SIZE	UNITS	CODE	DATE
Howe truss	10 nodes / 17 rods	m, kN, mm	CIRSOC 301	2026-10-12 01:58

λ is the factor on the load as it stands. $\lambda_1 = 1.90$, so the load can grow 1.9× before it buckles. 2 modes share that λ: that many equal members buckle at the same load, so stiffening one of them buys nothing.

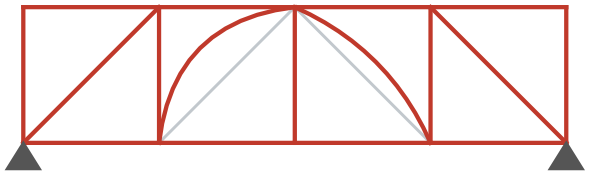
Mode 1: λ = 1.90
rod 13 bows

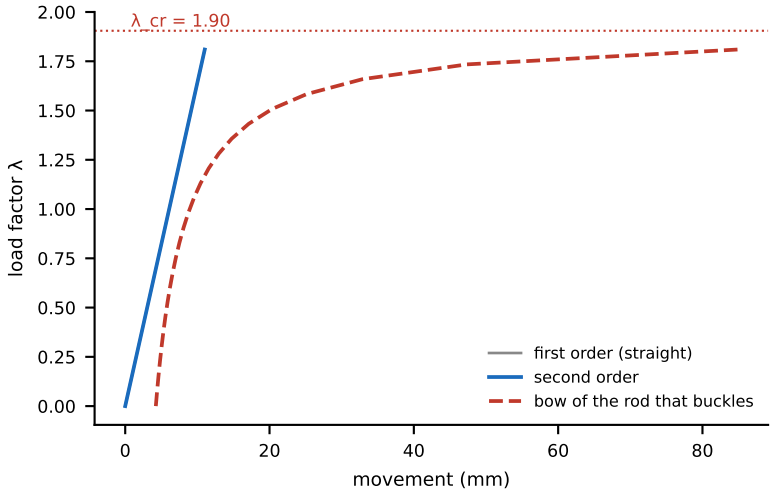


Mode 2: λ = 1.90
rod 16 bows



Mode 3: λ = 5.71
rod 14 bows





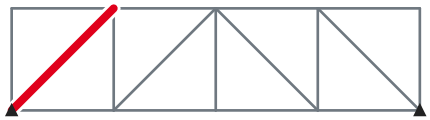
The second-order line bends away from the first-order one and runs off as λ nears λ_cr: that gap IS the amplification every second-order design rule is about. Started from an initial bow of L/1000 = 4.24 mm in the first mode's shape.

Elastic buckling of the IDEAL truss, so every λ is an UPPER bound: imperfections, residual stresses and yielding are not modelled, and each rod is cut in two so it can bow between its joints. Buckling OUT of the plane is the rod check's job (CIRSOC 301 E3, minor radius) and is in the Utilisation and Rods sheets, not here. A mode has a shape, not a size: the shapes are drawn exaggerated.

Truss — Buckling

MODEL	SIZE	UNITS	CODE	DATE
Howe truss	10 nodes / 17 rods	m, kN, mm	CIRSOC 301	2026-10-12 01:58

Rod	Nodes	Family	Section	L (m)	N (kN)	Governing check	Use	
0	0–1	Chord	CHS 114.3x4.5	3.00	+60.00	normal stress (H.3.4)	0.18	OK
1	1–2	Chord	CHS 114.3x4.5	3.00	+80.00	normal stress (H.3.4)	0.24	OK
2	2–3	Chord	CHS 114.3x4.5	3.00	+80.00	normal stress (H.3.4)	0.24	OK
3	3–4	Chord	CHS 114.3x4.5	3.00	+60.00	normal stress (H.3.4)	0.18	OK
4	5–6	Chord	CHS 114.3x4.5	3.00	+0.00	inelastic buckling (E3-2)	0.00	OK
5	6–7	Chord	CHS 114.3x4.5	3.00	-60.00	inelastic buckling (E3-2)	0.25	OK
6	7–8	Chord	CHS 114.3x4.5	3.00	-60.00	inelastic buckling (E3-2)	0.25	OK
7	8–9	Chord	CHS 114.3x4.5	3.00	+0.00	inelastic buckling (E3-2)	0.00	OK
8	0–5	Vertical	CHS 60.3x3.6	3.00	+0.00	normal stress (H.3.4)	0.00	OK
9	1–6	Vertical	CHS 60.3x3.6	3.00	+20.00	normal stress (H.3.4)	0.15	OK
10	2–7	Vertical	CHS 60.3x3.6	3.00	+0.00	normal stress (H.3.4)	0.00	OK
11	3–8	Vertical	CHS 60.3x3.6	3.00	+20.00	normal stress (H.3.4)	0.15	OK
12	4–9	Vertical	CHS 60.3x3.6	3.00	+0.00	normal stress (H.3.4)	0.00	OK
13	0–6	Diagonal	CHS 101.6x4.0	4.24	-84.85	inelastic buckling (E3-2)	0.69	OK
14	1–7	Diagonal	CHS 101.6x4.0	4.24	-28.28	inelastic buckling (E3-2)	0.23	OK
15	3–7	Diagonal	CHS 101.6x4.0	4.24	-28.28	inelastic buckling (E3-2)	0.23	OK
16	4–8	Diagonal	CHS 101.6x4.0	4.24	-84.85	inelastic buckling (E3-2)	0.69	OK



Rod 13

utilisation 0.69

inelastic buckling (E3-2)

```
Rod 13
======
Steel, CIRSOC 301-2018 -- A = 1,226 mm², Fy = 235 MPa, L = 4.243 m

 1. Slenderness  [E2]
      K·L / r
      = 1 × 4,243 mm / 34.5 mm
      = 122.8

 2. Euler stress  [E3-4]
      Fe = π² E / (KL/r)²
      = π² × 200,000 / 122.8²
      = 130.80 MPa

 3. Which branch  [E3]
      KL/r ≤ 4.71 √(E/Fy) ?
      = 122.8 vs 4.71 √(200,000/235) = 137.4
      = inelastic

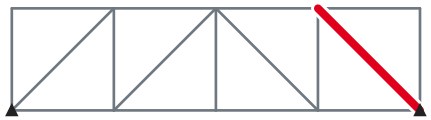
 4. Critical stress  [E3-2]
      Fcr = 0.658^(Fy/Fe) · Fy
      = 0.658^(235/130.80) × 235
      = 110.78 MPa

 5. Design strength  [E3-1]
      Pd = φc · Fcr · A
      = 0.9 × 110.78 × 1,226
      = 122.29 kN

 6. Compression ratio  [E3]
      |N| / Pd
      = 84.85 / 122.29
      = 0.694

 7. Governs
      the largest ratio
      = inelastic buckling (E3-2)
      = 0.694

utilisation 0.694 -- OK
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Rod 16

utilisation 0.69

inelastic buckling (E3-2)

```
Rod 16
======
Steel, CIRSOC 301-2018 -- A = 1,226 mm², Fy = 235 MPa, L = 4.243 m

 1. Slenderness  [E2]
      K·L / r
      = 1 × 4,243 mm / 34.5 mm
      = 122.8

 2. Euler stress  [E3-4]
      Fe = π² E / (KL/r)²
      = π² × 200,000 / 122.8²
      = 130.80 MPa

 3. Which branch  [E3]
      KL/r ≤ 4.71 √(E/Fy) ?
      = 122.8 vs 4.71 √(200,000/235) = 137.4
      = inelastic

 4. Critical stress  [E3-2]
      Fcr = 0.658^(Fy/Fe) · Fy
      = 0.658^(235/130.80) × 235
      = 110.78 MPa

 5. Design strength  [E3-1]
      Pd = φc · Fcr · A
      = 0.9 × 110.78 × 1,226
      = 122.29 kN

 6. Compression ratio  [E3]
      |N| / Pd
      = 84.85 / 122.29
      = 0.694

 7. Governs
      the largest ratio
      = inelastic buckling (E3-2)
      = 0.694

utilisation 0.694 -- OK
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